

Heterogeneous Verification for Multi-Step Decentralized Workflows

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Abstract

Financial and confidential automation relies on trusted intermediaries, imposing fees and custody risks that limit practical applications. Blockchains verify onchain operations but cannot verify automation requiring confidential execution, external data, or multi-step coordination. We propose a verification system where each operation uses the mechanism appropriate to its computation. The system combines cryptographic proofs for blockchain operations, hardware attestations for confidential execution, and stake-backed attestations for external interactions into workflow attestations. Validators broadcast attestations backed by stake, maintaining verifiability as long as honest validators control sufficient stake to make fraud economically irrational. This enables verifiable automation for financial operations, multi-party workflows, and confidential computation requiring trusted intermediaries.

1 Introduction

Almost all financial, legal, and confidential operations rely on trusted intermediaries. While these intermediaries work well enough for simple cases, they suffer from the inherent vulnerabilities of the trust-based model. Executing a will requires verifying death records through vital statistics offices, coordinating beneficiaries through legal counsel, transferring property titles through county registries, and distributing assets through financial institutions—operations that courts and executors have coordinated for centuries, yet disputes regularly destroy families. Medical records expose patient data to cloud providers, proprietary calculations depend on vendors, and sensitive operations leak to platforms. Algorithmic trading requires balance verification through blockchain systems, strategy execution in confidential environments, and trade settlement through exchanges that routinely fail through hacks and bankruptcy. With these dependencies comes concentration of power. Organizations and individuals must accept vendor lock-in and platform rules. A certain percentage of custody losses, contested estates, data breaches, and censorship is accepted as unavoidable. These risks can be avoided through manual in-person verification, but no mechanism exists for trustless execution at scale without centralized operators.

What is needed is a verification system based on cryptographic proof instead of trust, allowing processes to execute across multiple steps and parties without centralized intermediaries. We propose a system where validators verify each step and stake value backing their claims. Anyone can challenge fraudulent claims with proof. The system is secure as long as honest validators control enough stake to make fraud unprofitable.

2 Trust

No single mechanism can verify all these operations. Blockchain transactions are publicly verifiable through cryptographic proofs. Confidential computing uses sealed hardware to hide computation while proving it executed correctly. Courts, governments, and banks wield authority that cannot be verified—only trusted or feared.

References

- [1] S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” 2008.